

Marks Breakdown:

COs	Category	Criteria	Total Marks
CO3	Lab Performance	Assignment 01 = 4 marks Marking rubrics: 1 mark per feature based on problem identification, clarification and specification	10 (Total 32 Marks will be converted to 10)
		Assignment 02 = 4 marks Marking rubrics: 2 marks per feature (mapped to interfaces) based on proper navigation, input design, output design.	
		Assignment 03 = 4 marks Marking rubrics: 2 marks per feature (mapped to APIs). It should be connected to DB.	
		4 Features implementation. 5 marks per feature = 20 marks Features will be evaluated through module evaluation. Marking rubrics for each feature: Frontend (1) + Backend/API (1) + Database (1) + Innovation (1) + Integration with other features (1). Short viva can be taken for this evaluation. There is no late submission policy. The Jira project will be checked in every module.	
CO4	Presentation	Individual evaluation by project demo followed by viva	5
CO4	Report	Overall Structure = 2 Marks	5
		System Request = 1 Mark	
		User Manual = 1 Mark	
		Performance and Network Analysis = 1 Mark	
CO5	Project Simulation	Individual evaluation by live modification tasks on project (frontend, backend and database schema modification)	8
CO6	Project Management Tools	Github contribution by viva = 1 Mark	2
		Deployment contribution by viva = 1 Mark	
Total			30

Project Guideline

- Each group will have Minimum 3 members or Maximum 4 members.
- Each Member will implement a total of 4 features. Each member has to demonstrate his/her
 - 1st feature in module 1
 - 2nd feature in module 2 and
 - 3rd and 4th feature in module 3
- Project features will be evaluated in 3 modules: lab 5 (Module 1), lab 6 (Module 2) and lab 7 (Module 3). If any group member is absent/unable to demonstrate his/her feature(s) - *he/she will get zero marks on that particular feature.*
- **Login (Sign-On (SSO) / OAuth Login)+Signup+Logout, Role Management, Profile Management, Admin Activities will not be considered as features.** These will be considered common workflows for every project which can be implemented collaboratively.
- Each member must complete at least one external API integration to the system as a feature. For example: SMS Notifications, Push Notifications, Online Payments, Google Maps API, Google Directions API, Google Vision, Google Calendar API, Facebook Graph API, OpenAI, Google NLP, HuggingFace, etc.
- All web/desktop/android frameworks are allowed except **CMS (WordPress, Drupal, Joomla, Wix, Squarespace, and Shopify), core PHP, Firebase, Supabase, Django.**
- Students are permitted to use AI-assisted tools such as GPT, Claude, Gemini, GitHub Copilot, and similar platforms strictly for partial code generation and debugging support. However, please note the following:
 - AI dependency without understanding is strictly prohibited.
 - **Failure to properly explain any AI-generated code when asked will result in zero marks.**
 - Inability to perform live modifications (both frontend and backend) during evaluation will also lead to zero marks, regardless of overall project quality.
 - All students must ensure they maintain complete ownership and understanding of their codebase. AI may assist you but it cannot replace your logic, reasoning, or accountability.
- **UI/UX Requirement:** You have to follow Basic front-end layout standards.
 - It must be Responsive and mobile-friendly.
 - Allowed: Bootstrap, Tailwind CSS, Material UI, DaisyUI, Chakra UI, Ant Design, or any CSS library/framework that is *not a full CMS or page builder.*
 - Not Allowed: **Drag-and-drop builders like Elementor, Wix-style editors, or AI-generated full page templates without understanding the code structure.**
- **Database Requirement:** Must use SQL / NoSQL / Local Storage
- **Version Control Rule:** Each team must use GitHub/GitLab/Bitbucket with individual commit history. Copy-paste code without commits will be penalized.
- **Hardcode data is not accepted.** CRUD operations must be real.

FrameWork

Suggested Frameworks
PHP: Laravel, CodeIgniter
Java: Springboot, JavaFX
Javascript: MEAN, MERN, MEXN, NextJS
Python: FLASK, FastAPI
Android: Flutter, React Native, Kotlin
IOS: Flutter, React Native, SwiftUI
C#: ASP . NET CORE
Raw PHP, Django and Content Management System (CMS) like WordPress, Wix etc are not allowed.